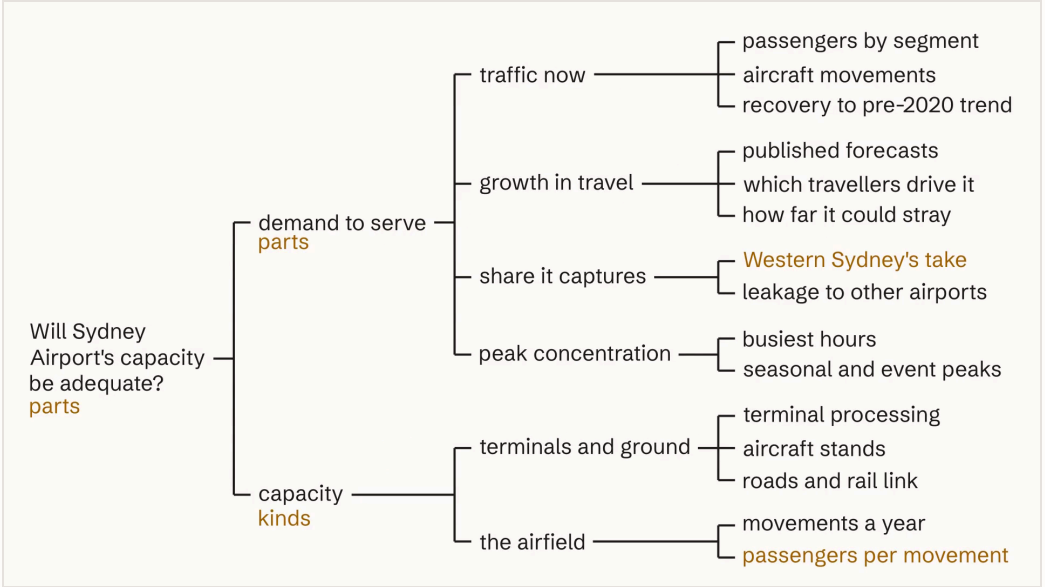
Open Agent Hackathon 2026 · Track 02 · The Agent That Can Explain Why · Jack Riddle

Descartes

A consulting team's method, run by agents: it breaks a hard business question into an issue tree, staffs every branch, and holds every figure to the passage it came from.

"Accept nothing as true that you cannot verify; divide every difficulty into as many parts as it takes." Descartes, *Discourse on the Method*, 1637 (paraphrased)

github.com/riddlejack/descartes



A real tree from a framing run, as drawn in the demo film: each fork splits its parent one way

The problem and who has it

Research with real breadth and depth takes a team of people, or gets done badly with deep-research reports

A$97,587

refunded by Deloitte on an A$440,000 government report that cited papers that don't exist (Oct 2025)

2,225

court cases in which AI-invented material reached a filing, 879 of them filed by lawyers (AI Hallucination Cases database, updated 27 Oct 2026)

For anyone who has to stand behind an answer

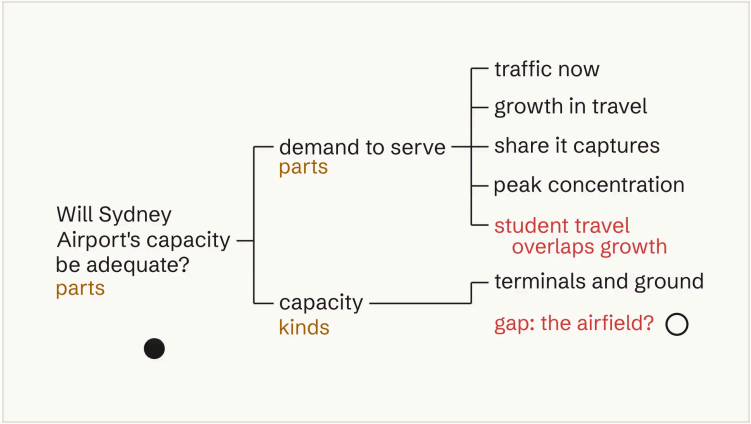
- Consultants, investment bankers, equity and credit analysts, strategy teams, businesses, scientists.
- And the reader who decides (a partner, a committee, a board), who can open any figure and see its sentence.
- Questions like: Should we enter this market? How big is it? Can this company pay its debts?

A deep-research tool gives a confident report in minutes, and nothing tells you what it skipped or whether its sources say what it claims.

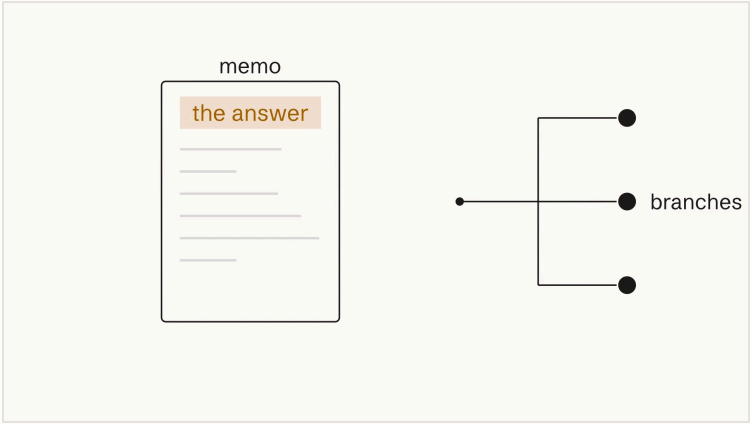
The solution

Descartes does both halves of a consulting team's job: the structure, and standing behind every number

- **The method.** Criteria the user confirms; a working answer tested against its strongest rival; an issue tree where each level splits its parent one way (parts, causes or kinds), in the discipline of Minto's *Pyramid Principle*; a blind second agent's critique; every branch staffed and reviewed.
- **The checks.** Every source captured and fingerprinted before it can be cited; a figure enters only by quoting its stored passage; code finds the quote and re-runs every calculation; an agent that didn't write the claim rules on it; Nemotron gives a second opinion.
- **The output.** A memo that answers the question first; every figure opens its passage; anything unchecked is listed. Code, not a model, decides release; anyone can re-check offline with no keys.



A second agent flags an overlap and a gap; the lead must answer each point



Final review: points go back to the owning branch; the memo leads with the answer

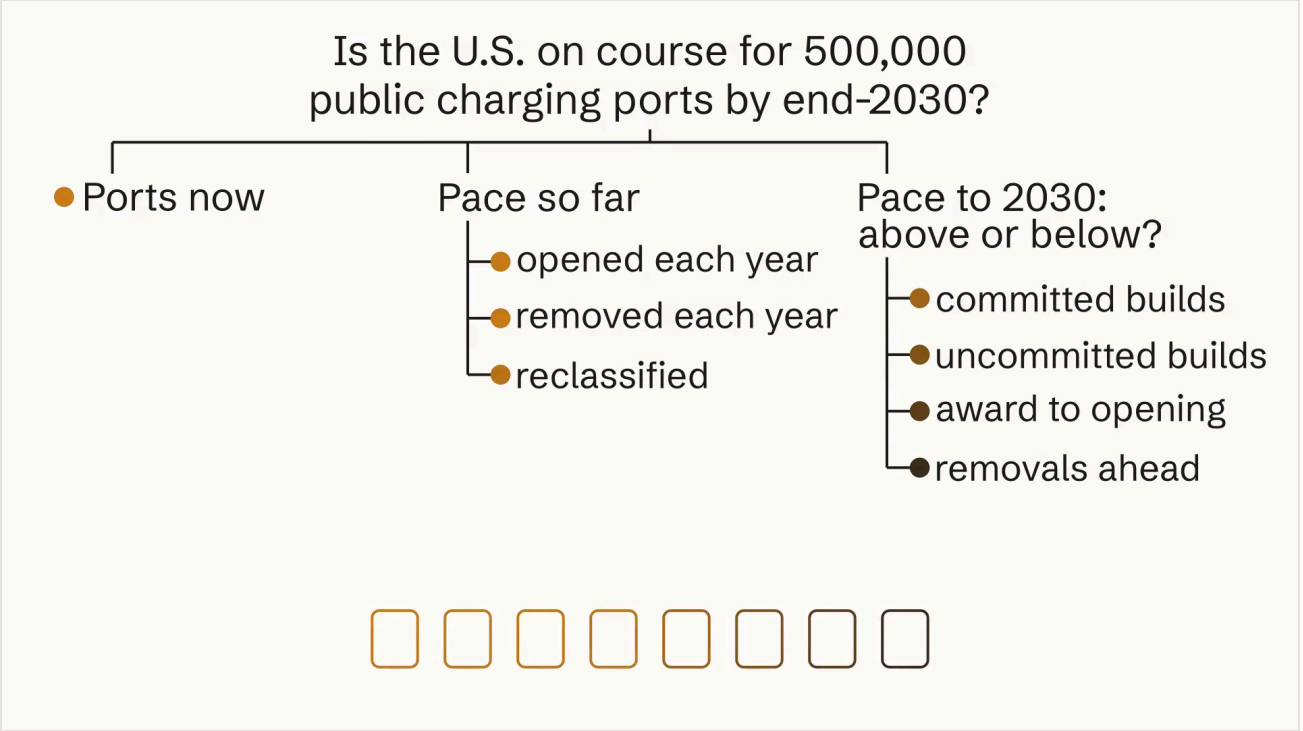
Method sources: Minto; Conn and McLean; Rasiel and Friga; Friga; Howson; decision research; Anthropic's multi-agent guidance. No author endorses this project.

The question becomes criteria, then an issue tree, then a blind critique the lead must answer

Criteria (from the question and the intake)

- Answer whether the U.S. is on track to reach 500,000 public EV charging ports by 2030.
- A short research answer, at the detail of a short memo.
- Let the person who asked know the answer, with no recommendation they did not ask for.

The blind framer, who never saw the lead's instructions, framed the question alone, then critiqued the tree. The lead answered every point: it restated the root, cut the 2030 subtree from 27 live questions to 17, and made two cruxes: whether the pace of additions is steady or accelerating, and whether openings fell after the 30C tax credit ended. Merged questions stay in the tree, marked set aside with the reason.



The real tree from the demo film, labels shortened to fit; merged questions not drawn

Framing run, 27 October 2026: 38 nodes, 13 set aside after the critique, two marked as cruxes.

Workflow, step 2 of 3

Each branch is staffed like an engagement, and a separate reviewer decides when it's answered

Each live question of the tree goes to a branch, and each branch gets its own researcher.

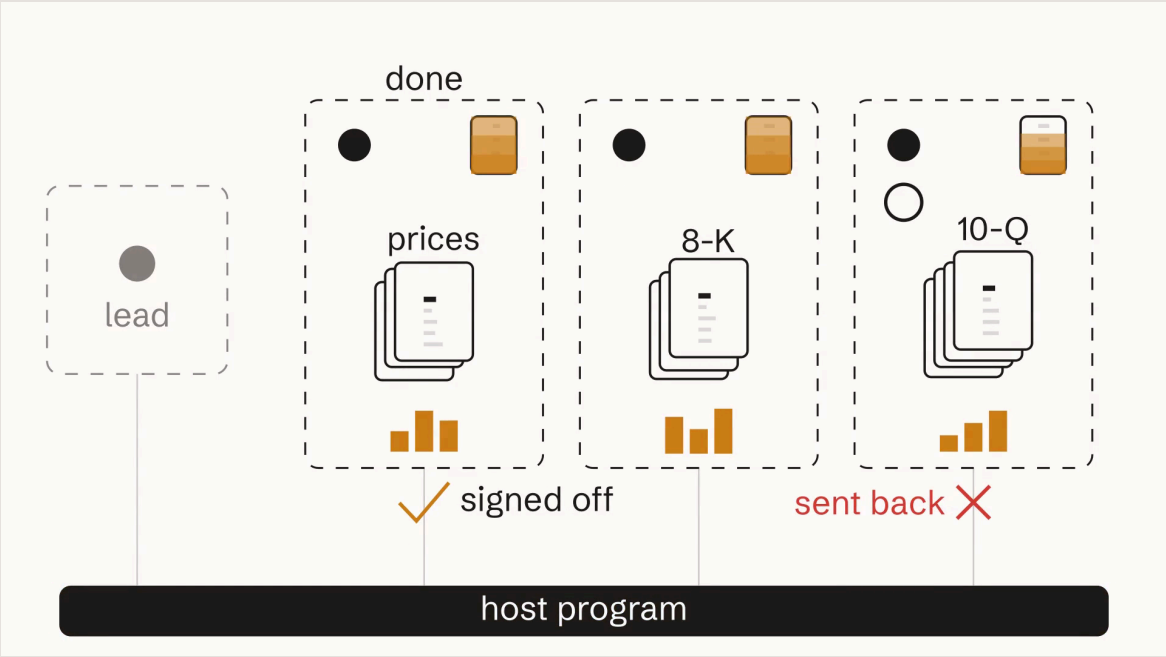


Illustration from the demo film: the host, not the lead, calls the reviewer

1. **Researcher (worker)** captures web pages, SEC filings and market data; files every figure as a claim quoting its passage; runs calculations from filed inputs
2. **Branch reviewer** a separate agent: is the branch's question really answered against the criteria? Closes it, or sends it back with numbered points
3. **Researcher revises** answers every point: resolves it, states it as a bound, or leaves it disputed for the lead
4. **Lead** reads branch records, never sources; connects what the branches found; writes the memo answering the question first

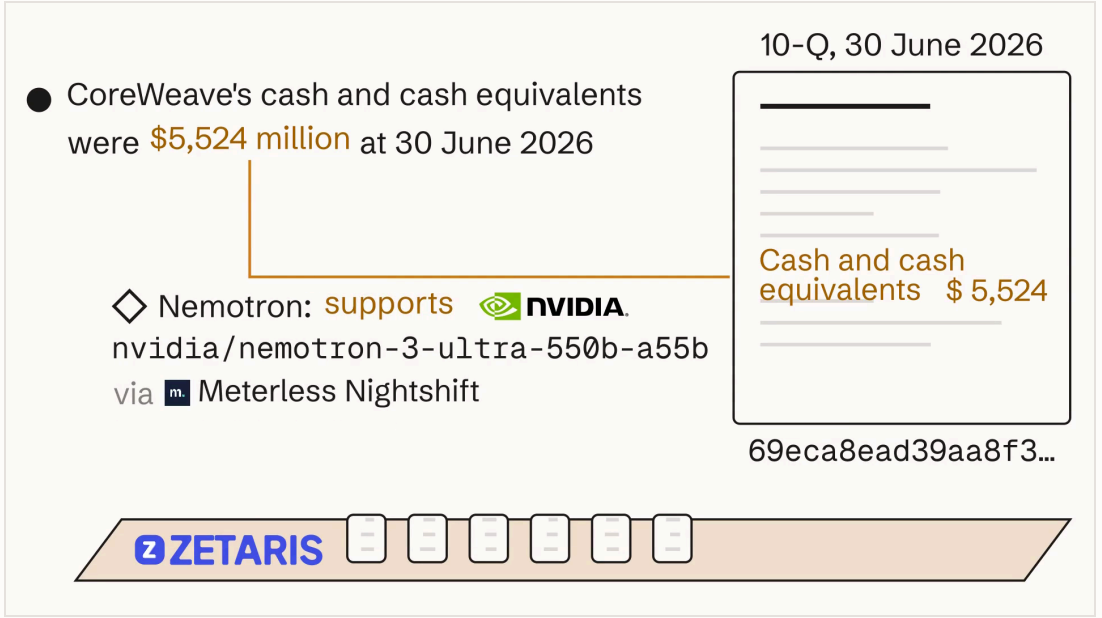
Release refuses while any branch is open. The lead cites only filed claims, so nothing reaches the memo except through a branch.

docs/ROLE-MAP.md (worker, branch-reviewer, lead)

Workflow, step 3 of 3 · the evidence loop

Every figure quotes a stored passage, and someone other than its author checks it

1. **Capture and fingerprint.** A page can be cited only after its bytes and text are stored under their SHA-256.
2. **Zetaris.** The run's evidence is one Unified Semantic Layer; every query logged with its SQL.
3. **Code checks.** The quotation is found in the stored text; each calculation's inputs are read back through Zetaris and it re-runs to exactly the same number.
4. **Claim checker.** An agent that is never the author reads the whole source and rules.
5. **Nemotron through Meterless Nightshift.** A second opinion from another model family, with a receipt naming the model; a disagreement starts a fresh checker.
6. **Meterless World Model register.** Every claim, verdict and change is recorded on World Model's data model.
7. **Release gate, in code.** No unsupported figure, no open branch, no open final review; anything unchecked is shown as "not checked".



A real claim from a real run, as shown in the demo film

On the first live run, Nemotron disagreed on 9 of 49 cited claims; a third look held all 9, and in 7 the quoted passage alone didn't show the claim, which is what a reader sees on opening it.

Technical stack

Python, Claude Code agents, and an evidence server any harness can speak to

Layer	What runs
Agents	Claude Opus 5.5 in Claude Code (pinned CLI 2.1.308), launched clean by a Python driver; role texts as a Claude Code plugin; hooks guard every tool call
Evidence service	An MCP server: capture with markdownify and BeautifulSoup (HTML), pypdfium2 (PDF text), edgartools (SEC filings, date-bounded), openpyxl and xlrd (spreadsheets); SHA-256 everywhere
Register	Meterless World Model's data model in Python; World Model's conformance runner in Node
Data layer	Zetaris (local Docker) over Postgres 16, or SQLite
Second family	NVIDIA Nemotron 3 Ultra and 3.5 Lightning through Meterless Nightshift's OpenAI-compatible Runtime
Output and checks	One static HTML report; `verify`, `tamper`, sample mode and Docker; Python 3.12, mypy strict, Apache-2.0
Built with	Claude Code (Claude Opus 5.5), with Codex as reviewer at merge gates; Cursor was never used

How Zetaris and Meterless were used, and what was unique

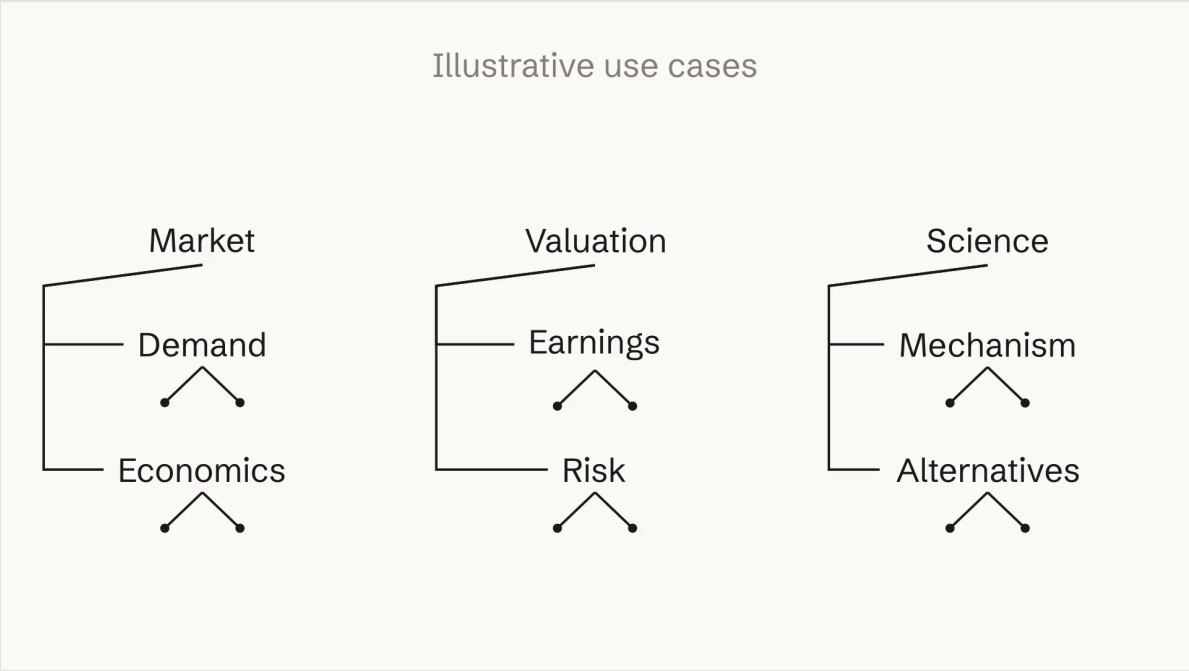
The sponsors sit inside the checking loop, each with a test result

Sponsor	What it does in Descartes	Test
ZETARIS Zetaris	One Unified Semantic Layer per run over the evidence; every calculation input read back through it, a one-bit difference refused; read-only `query_data`; every query logged	Live sync reads every value back bit for bit
m. Meterless World Model	The claim register and issue tree, on World Model's data model	Its own conformance suite, unmodified: 8 of 8; three deliberate breakages fail it
NVIDIA. Meterless Nightshift + NVIDIA Nemotron	Second opinion from another model family on each quoted claim; a receipt per call names the model	Run 1: 49 asked, 9 disagreed, all 9 held on a third look, 7 too-narrow quotes

Unique: a figure traces to a stored passage and a query a judge can see; the register that decides what was claimed is tested against World Model's own suite; a reviewer from another model family sits on every cited claim; and the question is Track 02's own shape. **Disclosure:** the research agents run on Claude in Claude Code; inside a run the host's own model calls go through Nightshift. Running the agents end to end through Nightshift needs a format bridge and API billing, not built.

Where it applies

Any question that has to be broken down before it can be answered

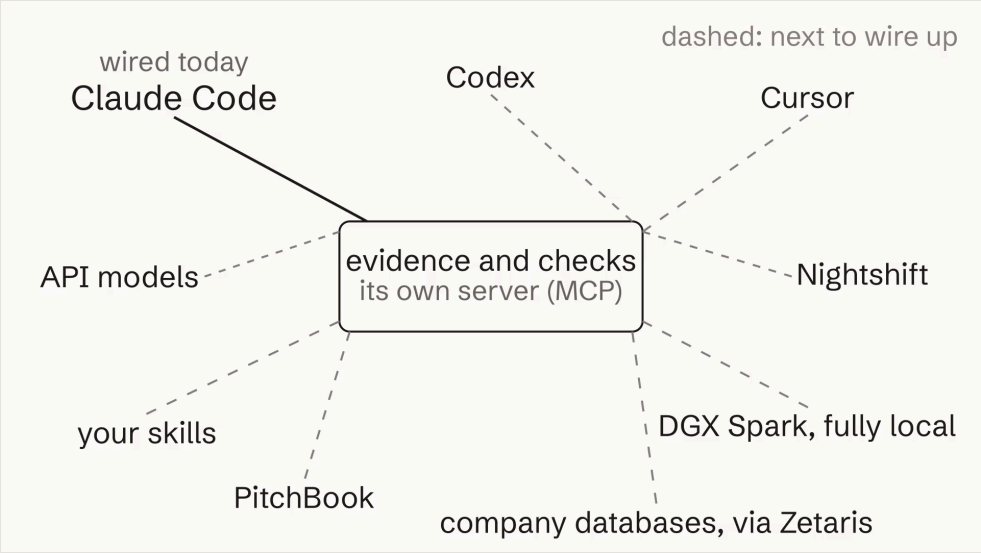


Illustrative use cases from the demo film

- Market entry and sizing, for a consultant or a business.
- Valuation, credit and hold-or-sell calls, for analysts and bankers.
- Mechanism and alternatives questions, in science.
- Not for quick look-ups, or where a mostly-right five-minute report is enough.

Future enhancements

Wired for Claude Code today; built to be driven from elsewhere



Solid: wired today. Dashed: next to wire up.

Next	Still to build
Codex, Cursor, API models, Nightshift end to end	Harness adapters (the evidence server is already MCP); a format bridge for Nightshift
Fully local, on a DGX Spark	Research on local models (untested)
Company databases through Zetaris	A company's query results captured as receipted evidence
PitchBook and other data	Paid-database connectors (SEC filings and market series already issue receipts)
Your own skills	Pinned card loading (four cards ship as skills; loading built on a branch)

Limits, plainly: a run takes hours; no run has yet released an answer end to end, and no accuracy comparison is claimed; a checker confirms a passage says what a claim says, not that the passage is true.